

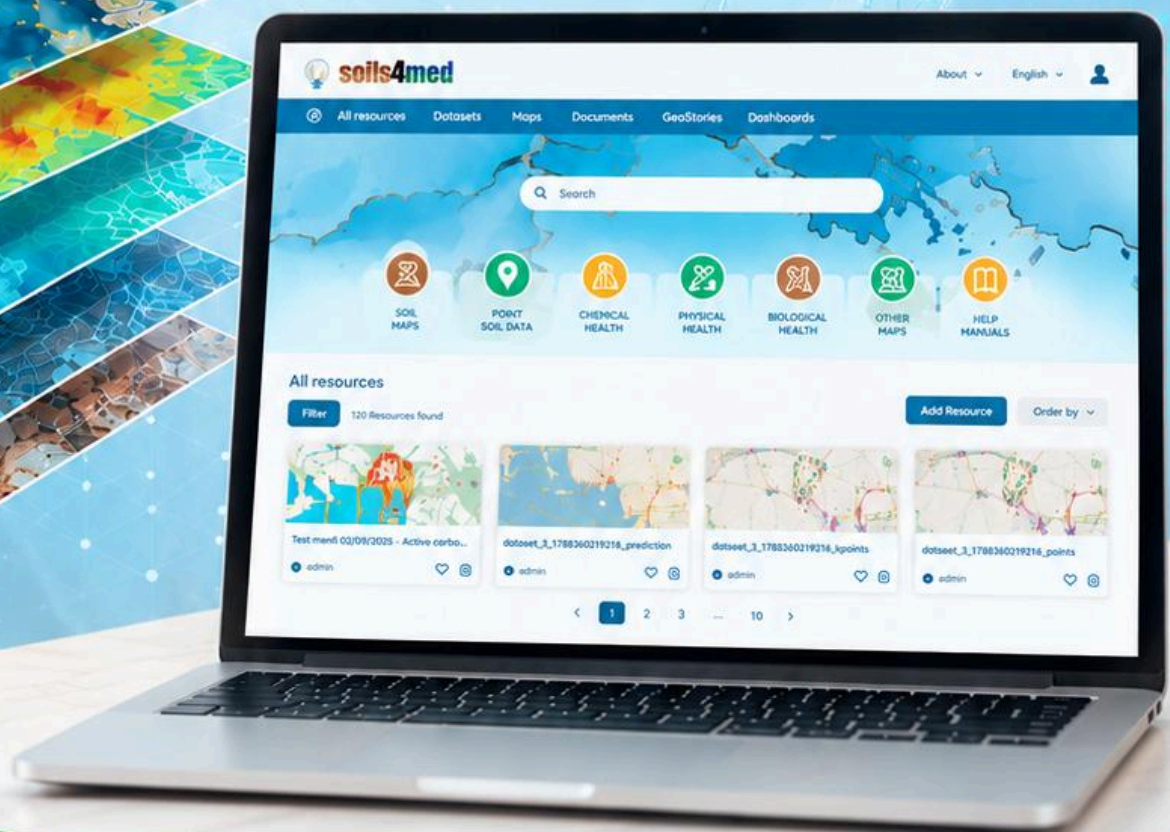
soils4med

SOIL health monitoring and information systems FOR
sustainable soil management in the MEDiterranean region

MedSIS manual

Soil Indicators

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DATASETS



MAPS



DOCUMENTS



GEOSTORIES



DASHBOARDS

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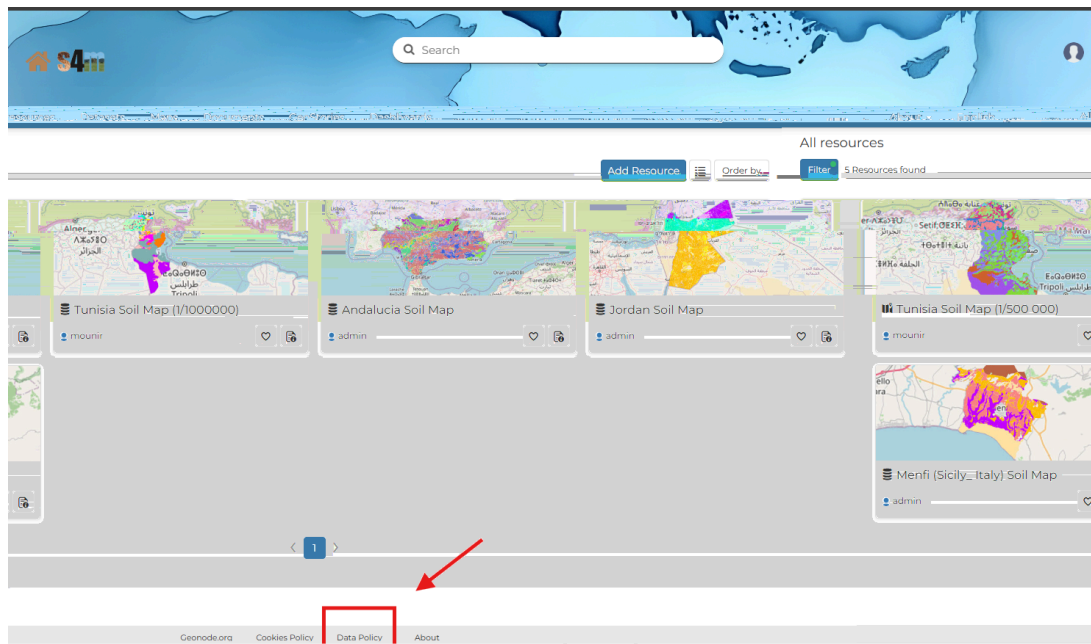
Overview

Formula

Short Abstract

Soil indicators - How To access to data

To access the soil data, the user must be registered in the system and submit a request to the data managers—specifically, the SIS staff—using the contact details provided on the "Data Policy" page shown in Figure 1.



Soil indicator name

area of Interest

period of observations

depth

type :

project

method

method

Identifier	Code	Description
SURVEY_METHODS:CP	CP	method: CPCS;
SURVEY_METHODS:FA	FA	method: FAO
SURVEY_METHODS:FA06	FA06	method: FAO; year: 2006
SURVEY_METHODS:FA52	FA52	method: FAO; year: 1952
SURVEY_METHODS:FA77	FA77	method: FAO; year: 1977
SURVEY_METHODS:FA90	FA90	method: FAO; year: 1990
SURVEY_METHODS:US	US	method: USDA soil survey manual
SURVEY_METHODS:US18	US18	method: USDA soil survey manual ; year: 2018
SURVEY_METHODS:US52	US52	method: USDA soil survey manual ; year: 1952
SURVEY_METHODS:US62	US62	method: USDA soil survey manual ; year: 1962
SURVEY_METHODS:US75	US75	method: USDA soil survey manual ; year: 1975
SURVEY_METHODS:US93	US93	method: USDA soil survey manual ; year: 1993
SURVEY_METHODS:WR	WR	method: WRB;
SURVEY_METHODS:WR22	WR22	method: WRB; year: 2022

ordinary kriging

List of soil indicators

Active carbonate (CaCo3)

%

Legend and value threshold:

☒ Active carbonate (CaCo3)

☐ No risk

☐ Potential iron chlorosis risk for sensitive crops (e.g. citrus species and Quercus suber)

☐ Potential iron chlorosis risk

☐ High iron chlorosis risk

label	rule
No risk	
Potential iron chlorosis risk for sensitive crops (e.g. citrus species and Quercus suber)	
Potential iron chlorosis risk	
High iron chlorosis risk	

SIS code name

Short abstract

%

Measure unit %

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	

<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>method</u>	" "
<u>geom</u>	

Antimony (Sb)

Legend and value threshold:

☒ Antimony (Sb)

☐ Normal

☐ Potential enrichment (Verify with EFSb)

label	rule
-------	------

Normal	
Potential enrichment (Verify with EFSb)	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "

<u>method</u>	" "
<u>geom</u>	

Arsenic (As)

Legend and value threshold:

☒ Arsenic (As)

☐ Normal

☐ Potential enrichment (Verify with EFAs)

label	rule
Normal	
Potential enrichment (Verify with EFAs)	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "

<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "
<u>method</u>	" " "
<u>geom</u>	

Available Phosphorus (P)

Legend and value threshold:

▼ Legend

☒ Available Phosphorus (P) content

- Too low (Olsen method only for cropland and grassland)
- Sufficient (Olsen method only for cropland and grassland)
- Medium (Olsen method only for cropland and grassland)
- High (Olsen method only for cropland and grassland)
- Very high (Olsen method only for cropland and grassland)

label	rule
Too low	

Sufficient

Medium

High

Very high

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name		description	
id			
point_id			
point_type		"	"
		"	"
date			
upper			
lower			
survey_m_id			

<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "
<u>method</u>	" " "
<u>geom</u>	

Base Saturation (BS) – Exchangeable Actidity

+ + + +

Legend and value threshold:

Legend

Base Saturation (BS) exchangeable acidity

Crops potentially affected by exchangeable acidity (for cropland only)

Crops potentially not affected by exchangeable acidity (for cropland only)

label	rule
Crops potentially affected by exchangeable acidity (for cropland only)	
Crops potentially not affected by exchangeable acidity (for cropland only)	

SIS code name

14

Short abstract

+ + + +

Measure unit %

Category

Involved SIS tables

Formula:

$$BS = \frac{ca_e + mg_e + k_e + na_e}{cec} \times 100$$

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	

	" " "
<u>geom</u>	

Base Saturation (BS) – Soil structure

+ + + +

Legend and value threshold:

☒ Base Saturation (BS) soil structure

Potential soil structure degradation risk

Lower potential soil structure degradation risk

label	rule
Potential soil structure degradation risk	
Lower potential soil structure degradation risk	

SIS code name

Short abstract

+ + + +

Measure unit %

Category

Involved SIS tables

Formula

$$BS = \frac{ca_e + mg_e + k_e + na_e}{cec} \times 100$$

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>geom</u>	

Bulk Density (BD)

Legend and value threshold:

-

label	rule
Suitable for plant growth	% % % %
Medium	% % % %
Affect root growth	% %

	% % %
Restrict root growth	% % % %

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

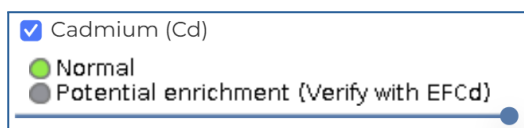
Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	

<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "
<u>method</u>	" " "
<u>clay</u>	%
<u>texture</u>	" " "
<u>geom</u>	

Cadmium (Cd)

Legend and value threshold:



label	rule
Normal	
Potential enrichment (Verify with EFCd)	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

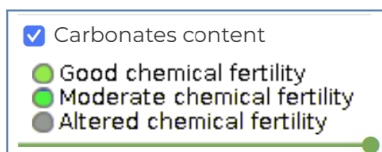
name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>method</u>	

	" "
<u>geom</u>	

Carbonate content

%

Legend and value threshold:



label	rule
Good chemical fertility	
Moderate chemical fertility	
Altered chemical fertility	

SIS code name

Short abstract %

Measure unit %

Category

Involved SIS tables

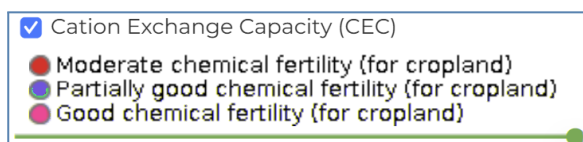
Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	

<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "
<u>method</u>	" "
<u>geom</u>	

Cation Exchange Capacity (CEC)

Legend and value threshold:



label	rule
Moderate chemical fertility	
Partially good chemical fertility	
Good chemical fertility	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>geom</u>	

Cation Exchange Capacity (CEC) / Clay ratio

Legend and value threshold:

- ☒ Cation Exchange Capacity (CEC) / Clay ratio
- Potential presence of low charge density clay
- Potential presence of high charge density clay with higher nutritive cations reservoir
- Potential presence of lignite, high charge density smectite or vermiculite

label	rule
Potential presence of low charge density clay	
Potential presence of high charge density clay	
Potential presence of lignite, high charge density smectite or vermiculite	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Formula

$$\frac{CEC}{Clay} = \frac{cec}{clay}$$

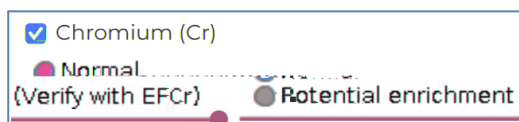
Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	

<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>geom</u>	

Chromium (Cr) content

Legend and value threshold:



label	rule
Normal	
Potential enrichment (Verify with EFCr)	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>method</u>	" "

W
geom
out

Cobalt (Co)



<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "
<u>method</u>	" " "
<u>geom</u>	

Copper (Cu)

Legend and value threshold:

☒ Copper (Cu)

☐ Normal

☐ Potential enrichment (Verify with EFCu)

label	rule
Normal	
Potential enrichment (Verify with EFCu)	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	

<u>unit</u>	" " "
<u>method</u>	" " "
<u>geom</u>	

Crust Cover

Legend and value threshold:

☒ Crust Cover ☐ Potentially more sensitive to wind erosion (for arid and desert environment) ☐ Potentially less sensitive to wind erosion (for arid and desert environment)

label	rule
Potentially more sensitive to wind erosion (for arid or semi-arid climate)	
Potentially less sensitive to wind erosion (for arid or semi-arid climate)	

SIS code name

Short abstract

Measure unit %

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>geom</u>	

Cryptogram cover

Legend and value threshold:

☒	Cryptogram Cover
●	No contribution to soil surface erosion protection
●	Slight contribution to soil surface erosion protection
●	Moderate contribution to soil surface erosion protection
●	Extensive contribution to soil surface erosion protection

label	rule
No contribution to soil surface erosion protection	
Slight contribution to soil surface erosion protection	
Moderate contribution to soil surface erosion protection	
Extensive contribution to soil surface erosion protection	

SIS code name

Short abstract

Measure unit %

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	

<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " " "
<u>geom</u>	

C/N ratio

Legend and value threshold:

☒ C/N ratio ☐ High SOC mineralization potential (Cropland and first layer only) ☐ Moderate SOC mineralization potential (Cropland and first layer only) ☐ Low SOC mineralization potential (Cropland and first layer only) ☐ High SOC mineralization potential and high microbial carbon accumulation activity (Grassland and forest with C/N less than 9.6 and 1st layer only) ☐ Moderate SOC mineralization potential and high microbial carbon accumulation activity (Grassland and forest with C/N less than 9.6 and 1st layer only) ☐ Low SOC mineralization potential and high microbial carbon accumulation activity (Grassland and forest with C/N less than 9.6 and 1st layer only) ☐ Moderate SOC mineralization potential and LOW microbial carbon accumulation activity (Grassland and forest with C/N is greater than or equal to 9.6 and 1st layer only) ☐ Low SOC mineralization potential and low microbial carbon accumulation activity (Grassland and forest with C/N is greater than or equal to 9.6 and 1st layer only)	
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label	rule
High SOC mineralization potential	
Moderate SOC mineralization potential	
Low SOC mineralization potential	
High SOC mineralization potential and higher microbial carbon accumulation activity	

Moderate SOC mineralization potential and higher microbial carbon accumulation activity	
Moderate SOC mineralization potential and lower microbial carbon accumulation activity	
Low SOC mineralization potential and lower microbial carbon accumulation activity	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Formula

$$\frac{C}{N} = \frac{org_car}{n_tot}$$

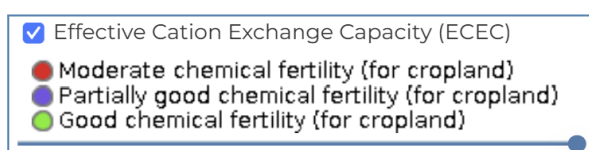
Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	

<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "
<u>geom</u>	

Effective Cation Exchange Capacity (ECEC)

Legend and value threshold:



label	rule
Moderate chemical fertility (for cropland)	
Partially good chemical fertility (for cropland)	
Good chemical fertility (for cropland)	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Formula

$$eCEC = \begin{cases} 1.02 * (ca_e + mg_e + na_e + k_e), & ph_h2o < 6.5 \\ 0.0143 * clay * 10 + 0.00237 * (ph_h2o - 4.5) * clay * 10 + 0.0488 * (ph_h2o - 1.2) * org_car, & ph_h2o \geq 6.5 \end{cases}$$

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>geom</u>	

Electric Conductivity

Legend and value threshold:

- ☒ Electric Conductivity (EC)
- ☐ Light to medium salinity
- ☐ Strong salinity
- ☐ Very strong salinity
- ☐ Excessive salinity

label	rule
Light salinity	$0.0 \leq x < 0.5$
Medium to strong salinity	$0.5 \leq x < 1.0$
Very strong salinity	$1.0 \leq x < 1.5$
Excessive salinity	$1.5 \leq x$

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
point_id	
point_type	"
date	
<u>upper</u>	



<u>survey m id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " " "
<u>method</u>	" " "
<u>geom</u>	

Exch. Ca (Ca++)

Legend and value threshold:

☒ Exchangeable Calcium (Ca++) cation

- Low
- Medium
- High
- Very high

label	rule
Low	
Medium	
High	
Very high	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey m id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>method</u>	" "
<u>geom</u>	

Exch. Mg⁺⁺

Legend and value threshold:

☒ Exchangeable Magnesium cation (Mg⁺⁺) cation

- Low
- Medium
- High
- Very high

label	rule
Low	
Medium	
High	
Very high	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

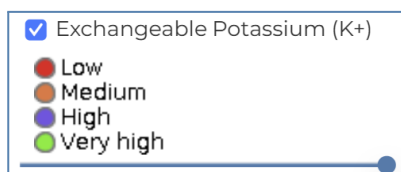
Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	

<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "
<u>method</u>	" " "
<u>geom</u>	

Exch. K+

Legend and value threshold:



label	rule
Low	
Medium	
High	
Very high	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>method</u>	" "
<u>geom</u>	

Exch. K⁺ impact on production

Legend and value threshold:

☒	Exchangeable Potassium (K ⁺) cation, potential impact on production
●	Potential positive impact on productivity following potassium fertilizers in sandy soils
●	Potential non positive impact on productivity following potassium fertilizers in sandy soils
▲	Potential positive impact on productivity following potassium fertilizers in clayey soils
▲	Potential non positive impact on productivity following potassium fertilizers in clayey soils

label	rule
Potential positive impact on productivity following potassium fertilizers in sandy soils	
Potential non positive impact on productivity following potassium fertilizers in sandy soils	
Potential positive impact on productivity following potassium fertilizers in clayey soils	
Potential non positive impact on productivity following potassium fertilizers in clayey soils	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "
<u>texture</u>	" " "
<u>geom</u>	

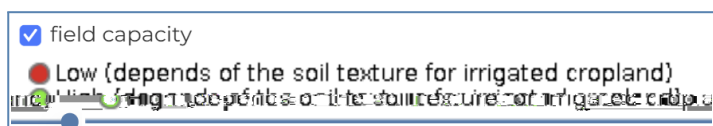
Exch. Na⁺



<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>method</u>	" "
<u>geom</u>	

Field Capacity

Legend and value threshold:



label	rule
-------	------

Low	
High	

SIS code name

Short abstract

Measure unit %

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "

<u>texture</u>	" "
<u>geom</u>	

Gravel

Legend and value threshold:

☒ gravel

Absent or low

Normal

Frequent

Abundant

label	rule
Absent or low	
Normal	
Frequent	
Abundant	

SIS code name
 Short abstract
 Measure unit %
 Category
 Involved SIS tables
 Indicator table fields

name	description
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Negative impact for plant growth (for cropland)	
---	--

SIS code name

Short abstract

Measure unit %

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	

<u>unit</u>	" " "
<u>geom</u>	

Hydraulic conductivity

Legend and value threshold:

☒ Hydraulic conductivity at saturation

Very low

Low

Moderate

High

Very high

label	rule
Very low	
Low	
Moderate	
High	
Very high	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>method</u>	" "
<u>geom</u>	

Pb

Legend and value threshold:

☒ Lead (Pb) mg/kg

☐ Normal

☐ Potential enrichment (Verify with EFPb)

label	rule
Normal	
Potential enrichment (Verify with EFPb)	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	

<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "
<u>method</u>	" "
<u>geom</u>	

RC_{Pb} - Lead relative content

Legend and value threshold:

☒ Lead relative content
☐ Lead below natural geochemical relative range
☐ Lead natural geochemical relative range
☐ Lead above natural geochemical relative range

label	rule
Lead below natural geochemical relative range	
Lead natural geochemical relative range	
Lead above natural geochemical relative range	

SIS code name

Short abstract

Measure unit %

Category

Involved SIS tables

Formula

$$RC_{Pb} - \frac{pb}{[cu + zn + pb]} * 100$$

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>geom</u>	

Litter layer

Legend and value threshold:



label	rule
Slight cover	
Moderate cover	
High cover	
Extensive cover:	

SIS code name

Short abstract

Measure unit %

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	

<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " " "
<u>geom</u>	

Manganese

Legend and value threshold:

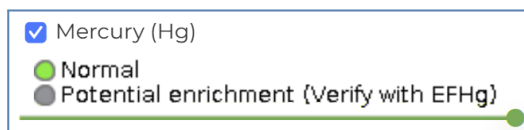
☒ Manganese (Mn)

☐ Normal

☐ Potential enrichment (Verify with EFMn)

label	rule
Normal	
Potential enrichment (Verify with EFMn):	

Legend and value threshold:



label	rule
Normal	
Potential enrichment (Verify with EFHg):	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

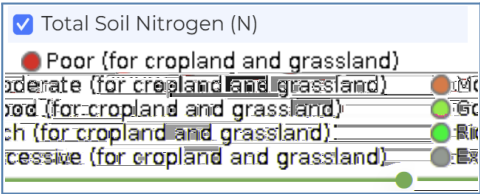
Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	

<u>unit</u>	" " "
<u>method</u>	" " "
<u>geom</u>	

N total content

Legend and value threshold:



label	rule
Poor	
Moderate	
Good	
Rich	
Excessive	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
------	-------------

<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>geom</u>	

NH₄⁺

Legend and value threshold:

☒ Ammonium ion (NH ₄ ⁺)
☐ Low range (for cropland)
☐ Normal range (for cropland)
☐ Wet soil (anaerobic) or residual fertilizer (for cropland)

label	rule
Low range (for cropland)	
Normal range (for cropland	
Wet soil (anaerobic) or residual fertilizer (for cropland)	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "

<u>geom</u>	
-------------	--

Ni

Legend and value threshold:

☒ Nichel (Ni)
☐ Normal
☐ Potential enrichment (Verify with EFNi)

label	rule
Normal	
Potential enrichment (Verify with EFNi)	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	

<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " " "
<u>method</u>	" "
<u>geom</u>	

NO₃-

Legend and value threshold:

☒ Nitrate ion (NO ₃ -)
☐ Low (for cropland)
☐ Medium (for cropland)
☐ High (for cropland)
☐ Excessive (leaching risk for cropland)

label	rule
Low (for cropland)	
Medium (for cropland)	
High (for cropland)	
Excessive (leaching risk) (for cropland)	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "

geom

Organic Carbon

Legend and value threshold:

- ☒ Organic Carbon (SOC)
- Very low SOC and low potential of soil ecosystem functions
 - Low SOC and low potential of soil ecosystem functions
 - High SOC and low potential of soil ecosystem functions
 - Very high SOC and low potential of soil ecosystem functions
 - Very high SOC and medium potential of soil ecosystem functions
 - ▲ Very high SOC and high potential of soil ecosystem functions

label	rule
Very low to low SOC and low potential of soil ecosystem services	
Medium SOC and low potential of soil ecosystem services	
High SOC and low potential of soil ecosystem services	
Very high SOC and low potential of soil ecosystem services	
Very high SOC and medium potential of soil ecosystem services	
Very high SOC and high potential of soil ecosystem services	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	

<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "
<u>method</u>	" "
<u>geom</u>	

Organic Matter

Legend and value threshold:



label	rule
Too low	
Low	
Medium	
Good	

SIS code name

Short abstract

Measure unit %

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	

<u>unit</u>	" " "
<u>method</u>	" " "
<u>geom</u>	

Oxygen availability

Legend and value threshold:

Legend

☒ Oxygen availability for roots

● Low (soil easily aerated optimal for most of plant growth)

● Medium (potential beginning of O2 turnover depletion and nitrate reduction rate potentially increase facilitating potentially root growth)

● Good (very low O2 turnover)

label	rule
Low (soil easily aerated optimal for most of plant growth)	
Medium (potential beginning of O2 turnover depletion and nitrate reduction rate potentially increase facilitating potentially root growth)	
Good (very low O2 turnover)	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Formula

$$\begin{cases} \frac{D}{D_0} = \left(\frac{satur - field_cap}{satur} \right)^{\frac{4}{3}}, & \text{if } satur = value \\ \frac{D}{D_0} = \left(\frac{((1 - \frac{bulk_dens}{2.65}) \times 100) - field_cap}{(1 - \frac{bulk_dens}{2.65}) \times 100} \right)^{\frac{4}{3}}, & \text{if } satur = NA \end{cases}$$

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "
<u>geom</u>	

Phosphorus Threshold

Legend and value threshold:

Legend

- ☒ Phosphorus Threshold for Plant growth
- Potential positive impact on productivity following phosphorus fertilizers
- Potential non responsive impact on productivity following phosphorus fertilizers
- Potential leaching risk

label	rule
Potential positive impact on productivity following phosphorus fertilizers	
Potential non responsive impact on productivity following phosphorus fertilizers	
Potential leaching risk	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Formula

$$\frac{P}{P_{threshold}} = \begin{cases} \frac{p_{cont}}{49 - 0.016 (clay * 10) - 3.81 \frac{ph_{h2o}}{19.4}}, & Land\ cover = Cropland \\ \frac{p_{cont}}{19.4}, & Land\ cover = Grassland \end{cases}$$

Indicator table fields

name	description
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<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "
<u>geom</u>	

Packing density

Legend and value threshold:

☒ Packing density

Low compaction

Medium compaction

High compaction

label	rule
Low compaction	
Medium compaction	
High compaction	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Formula

$$PD = bulk_dens + 0.009 * clay$$

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	

<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "
<u>geom</u>	

Perennial Vegetation Cover (PVC)

Legend and value threshold:

☒ Perennial vegetation cover
☐ No belowground contribution to soil surface erosion protection
☐ Low belowground contribution to soil surface erosion protection
☐ Moderate belowground contribution to soil surface erosion protection
☐ High belowground contribution to soil surface erosion protection

label	rule
No below ground contribution to soil surface erosion protection	
Low below ground contribution to soil surface erosion protection	
Moderate below ground contribution to soil surface erosion protection	
High below ground contribution to soil surface erosion protection	

SIS code name

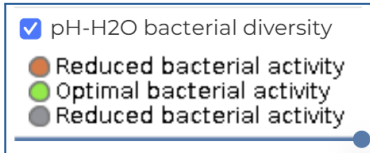
Short abstract



geom

pH-H2O bacterial diversity

Legend and value threshold:



label	rule
Reduced bacterial activity	
Optimal bacterial activity	
Reduced bacterial activity	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

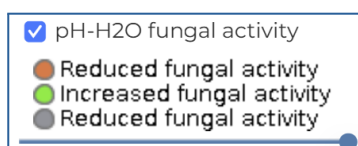
Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	

<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " " "
<u>method</u>	" " " "
<u>geom</u>	

pH-H₂O fungal activity

Legend and value threshold:



label	rule
Reduced fungal activity	
Increased fungal activity	
Reduced fungal activity	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "
<u>geom</u>	

pH-H2O metal toxicity

Legend and value threshold:

☒ pH-H2O metal toxicity

More toxicity due to heavy metals dissolution

Lower toxicity due to heavy metals dissolution

More toxicity due to heavy metals dissolution

label	rule

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " "

<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "
<u>geom</u>	

pH-H2O microbial carbon

Legend and value threshold:

☒ pH-H2O microbial carbon

☐ Reduced microbial carbon use efficiency (goes to 0)

☐ Higher microbial carbon use efficiency

label	rule
Reduced microbial carbon use efficiency (goes to 0)	
Higher microbial carbon use efficiency	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>geom</u>	

pH-H2O microbial diversity

Legend and value threshold:



label	rule
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Reduced microbial diversity	
Higher and more stable microbial diversity	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>method</u>	" "

<u>geom</u>	
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pH-H2O organic decomposition

Legend and value threshold:

☒ pH-H2O organic decomposition
☐ Slower organic matter decomposition
☐ Optimal organic matter decomposition
☐ Too high organic matter decomposition

label	rule
Slower organic matter decomposition	
Optimal organic matter decomposition	
Too high organic matter decomposition	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	

	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "
<u>method</u>	" " "
<u>geom</u>	

pH-H₂O phosphorus

Legend and value threshold:

☒ pH-H ₂ O phosphorus	
☐ Limited P cycling through acid phosphatase activity	
☐ Optimal phosphorus availability with acid phosphorus enzyme activity depending on available phosphorus soil stock	
☐ Limited phosphorus availability with acid phosphorus enzyme activity	

label	rule
Limited P cycling through acid phosphatase activity	

Optimal phosphorus availability with acid phosphorus enzyme activity depending on available phosphorus soil stock	
Limited phosphorus availability with acid phosphorus enzyme activity	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	

<u>value</u>	
<u>unit</u>	" " "
<u>geom</u>	

pH-H₂O plant growth

Legend and value threshold:

☒ pH-H ₂ O plant growth ☐ Fe and Al ³⁺ dissolution affecting relative yields (for cropland) ☐ Optimal plant growth least limiting range (for cropland) ☐ Plant growth limiting range (species specific) with increasing constraints regarding excess of HC03 ⁻ or deficiency potential in Fe, P and Zn or imbalance for and in between K, Mg and/or Ca (for cropland)
--

label	rule
Fe and Al ³⁺ dissolution affecting relative yields (for cropland)	
Optimal plant growth least limiting range (for cropland)	
Plant growth limiting range (species specific) with increasing constraints regarding excess of HC03 ⁻ or deficiency potential in Fe, P and Zn or imbalance for and in between K, Mg and/or Ca (for cropland)	

SIS code name

Short abstract

Measure unit

Category



label	rule
Low probability of soil sodicity and structural deterioration	
Higher probability of soil sodicity and structural deterioration	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>geom</u>	

pH-H2O som for grassland

Legend and value threshold:

☒ pH-H2O som for grassland
☐ Less redistribution of SOM for bioturbation by earthworm activity (specifically for grassland)
☐ Optimal redistribution of SOM for bioturbation by earthworm activity (specifically for grassland)
☐ Less redistribution of SOM for bioturbation by earthworm activity (specifically for grassland)

da fare

label	rule
Low redistribution of SOM for bioturbation by earthworm activity (specifically for grassland)	
Optimal redistribution of SOM for bioturbation by earthworm activity (specifically for grassland)	
Low redistribution of SOM for bioturbation by earthworm activity (specifically for grassland)	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	

<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " " "
<u>method</u>	" " "
<u>geom</u>	

Plant Available Water Capacity (AWC)

Legend and value threshold:

- ☒ Plant Available Water Capacity (AWC)
- ☐ Poor water storage capacity (for rainfed cropland and grassland)
- ☐ Limited water storage capacity (for rainfed cropland and grassland)
- ☐ Good water storage capacity (for rainfed cropland and grassland)
- ☐ Ideal water storage capacity (for rainfed cropland and grassland)

label	rule
Poor water storage capacity (for rainfed cropland and grassland)	
Limited water storage capacity (for rainfed cropland and grassland)	
Good water storage capacity (for rainfed cropland and grassland)	
Ideal water storage capacity (for rainfed cropland and grassland)	

SIS code name

Short abstract

Measure unit %

Category

Involved SIS tables

Formula

$$AWC-field_cap - wilting_p$$

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>method</u>	" "
<u>geom</u>	

Relative Field Capacity (RFC)

Legend and value threshold:

☒ Relative field capacity

Water limited soil {for rainfed cropland and grassland}

Optimal {for rainfed cropland and grassland}

Air limited soil {for rainfed cropland and grassland}

label	rule
Water limited soil (for rainfed cropland and grassland)	
Optimal (for rainfed cropland and grassland)	
Air limited soil (for rainfed cropland and grassland)	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Formula

$$RFC = \begin{cases} field_cap / satur, & \text{if } satur = present \\ \frac{field_cap}{1 - \left(1 - \frac{bulk_dens}{2.65}\right) * 100}, & \text{if } satur = NA \end{cases}$$

Indicator table fields

name	description
<u>id</u>	

<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>geom</u>	

Slake test (STAB10)

Legend and value threshold:

☒ Slake test

Unstable

Moderately stable

Stable

label	rule
Unstable	
Moderately stable	

Stable	
--------	--

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>method</u>	
<u>geom</u>	

SOC/clay

Legend and value threshold:

Legend

☒ SOC/Clay ratio

- Poor (for topsoil, in agricultural and grassland only)
- Potential improvement (for topsoil, in agricultural and grassland only)
- Good (for topsoil, in agricultural and grassland only)
- Very good (for topsoil, in agricultural and grassland only)

label	rule
Poor (for topsoil, in agricultural and grassland only)	
Potential improvement (for topsoil, in agricultural and grassland only)	
Good (for topsoil, in agricultural and grassland only)	
Very good (for topsoil, in agricultural and grassland only)	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Formula

$$\frac{SOC}{Clay} = \frac{\frac{org_{car}}{10}}{clay}$$

Indicator table fields

name	description
<u>id</u>	

point_id

point_type

"

"

"

date

upper

lower

survey_m_id

"

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project

horizon

value

unit

"

"

"

geom

Soil Organic Carbon (SOC) st



label	rule
Degraded equivalence (Only for grassland and cropland)	
Strong pressure equivalence (Only for grassland and cropland)	
Weak to moderate pressure equivalence (Only for grassland and cropland)	
Very weak to no pressure equivalence (Only for grassland and cropland)	
Low dense forests stands (Only for forest)	
Moderately dense to dense forests stands equivalent (Only for forest)	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Formula

$$SOC_{stock} = \frac{org_car}{10} * BD * (lim_down - lim_up) * \left(1 - \frac{gravel}{100}\right)$$

Indicator table fields

name	description
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<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>geom</u>	

Sodicity Salinity Ratio (SAR/EC)

Legend and value threshold:

☒ Sodicity Salinity Ratio

- Unbeneficial impact of gypsum input for salinity remediation (only for saturated paste extract)
- Beneficial impact of gypsum input for salinity remediation (only for saturated paste extract)

label	rule
Unbeneficial impact of gypsum input for salinity remediation (only for saturated paste extract)	
Beneficial impact of gypsum input for salinity remediation (only for saturated paste extract)	

SIS



	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " " "
<u>geom</u>	

Sodium Adsorption Ratio (SAR) sodicity

Legend and value threshold:

☒	Sodium Adsorption Ratio (SAR) sodicity
☐	Non sodium impacted soil
☐	Sodium impacted soil

label	rule
Non sodium impacted soil layer (only for saturated paste extract)	
Sodium impacted soil layer (salin soil) (only for saturated paste extract)	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Formula

$$SAR = \frac{na_{sol}}{(ca_{sol} + mg_{sol})^{\frac{1}{2}}}$$

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	

<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " " "
<u>geom</u>	

Sodium Adsorption Ratio (SAR) toxicity

Legend and value threshold:

☒	Sodium Adsorption Ratio (SAR) toxicity
☐	No toxicity for sensitive culture
☐	Restriction toxicity for sensitive culture
☐	Serious restriction toxicity for sensitive culture

label	rule
No toxicity for sensitive culture	
Restriction toxicity for sensitive culture	
Serious restriction toxicity for sensitive culture	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Formula

$$SAR = \frac{na_{sol}}{(ca_{sol} + mg_{sol})^{\frac{1}{2}}}$$

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>geom</u>	

Sodium Exchangeable Percentage (ESP) sodicity

Legend and value threshold:

☒ Sodium Exchangeable Percentage (ESP) sodicity

Soil non sodic

Soil sodic

label	rule
Soil non sodic	
Soil sodic	

SIS code name

Short abstract

Measure unit %

Category

Involved SIS tables

Formula

$$ESP=\frac{na-e}{cec} * 100$$

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	

<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " " "
<u>geom</u>	

Sodium Exchangeable Percentage (ESP) waterlogging

Legend and value threshold:

▼ Legend

☒ Sodium Exchangeable Percentage (ESP) waterlogging

- Lower waterlogging potential risk due to soil sodicity
- Higher waterlogging potential for soil with high charge density clay
- Higher waterlogging potential risk due to soil sodicity
- Lower waterlogging potential risk due to soil sodicity
- Higher waterlogging potential risk due to soil sodicity

label	rule
Lower waterlogging potential risk due to soil sodicity	
Higher waterlogging potential for soil with high charge density clay	
Higher waterlogging potential risk due to soil sodicity	
Lower waterlogging potential risk due to soil sodicity	
Higher waterlogging potential risk due to soil sodicity	

--	--

SIS code name

Short abstract

Measure unit %

Category

Involved SIS tables

Formula

$$ESP = \frac{na_e}{cec} * 100$$

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "

<u>cec clay ratio</u>	
<u>geom</u>	

Soil Erodibility by Water

Legend and value threshold:

☒ Soil Erodibility by Water
● Slight soil erodibility potential by water
● High soil erodibility potential by water

label	rule
Slight soil erodibility potential by water	
High soil erodibility potential by water	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Formula

$$K = \frac{\left[2.1 \times 10^{-4} \times \left(12 - \left(\frac{org_{car} \times 1.724}{10} \right) \right) \times [silt \times (silt + sand)]^{1.14} + 3.25(s-2) + 2.5(p-3) \right]}{100 \times 7.59}$$

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	

<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "
<u>geom</u>	

Soil Erodibility by Wind (WEF)

Legend and value threshold:

☒ Soil Erodibility by Wind
● Slightly erodible (for cropland and topsoil only)
● Moderately erodible (for cropland and topsoil only)
● Highly erodible (for cropland and topsoil only)
● Extremely erodible (for cropland and topsoil only)

label	rule
Slightly erodible (for cropland and topsoil only)	
Moderately erodible (for cropland and topsoil only)	
Highly erodible (for cropland and topsoil only)	

Extremely erodible (for cropland and topsoil only)	
--	--

SIS code name

Short abstract

Measure unit %

Category

Involved SIS tables

Formula

$$WEF = (-40.20 + 1.44 * sand + 1.08 * silt - 5.4 * \frac{sand}{clay} + 6.13 * \frac{org_{car}}{10} + 84.17 * caco3$$

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "

geom

Surface Compaction Risk

Legend and value threshold:

- ☒ Surface Compaction Risk
- Very low to low (for topsoil only on soil with less than 13% sand)
 - Medium (for topsoil only on soil with less than 13% sand)
 - High (for topsoil only on soil with less than 13% sand)
 - Very high (for topsoil only on soil with less than 13% sand)

label	rule
Very low to low (for topsoil only on soil with less than 13% sand)	
Medium (for topsoil only on soil with less than 13% sand)	
High (for topsoil only on soil with less than 13% sand)	
Very high (for topsoil only on soil with less than 13% sand)	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Formula

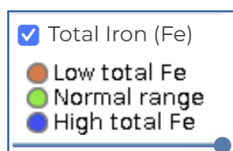
$$c = 0.95 + 0.016(\text{clay} + \text{silt}) - 0.00012(\text{clay} + \text{silt})^2 + e^{-(0.49+0.27*\text{org_mat})}$$

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>geom</u>	

Total Iron (Fe)

Legend and value threshold:



label

rule

Low total Fe	
Normal range	
High total Fe	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>method</u>	" "

<u>geom</u>	
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Vanadium (V)

Legend and value threshold:

☒ Vanadium (V)

☐ Normal

☐ Potential enrichment (Verify with EFV)

label	rule
Normal	
Potential enrichment (Verify with EFV)	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
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<u>survey_m_id</u>	" " " "
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<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " " "
<u>method</u>	" " "
<u>geom</u>	

Volumetric water content at wilting point

Legend and value threshold:

☒ Wilting Point
☐ Difficult conditions for plants in very dry condition
☐ Less difficult conditions for plants in very dry conditions

label	rule
Difficult conditions for plants in very dry condition	
Less difficult conditions for plants in very dry conditions	

SIS code name

Short abstract

Measure unit %

Category

Involved SIS tables

Indicator table fields

name	description
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<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>method</u>	" "
<u>geom</u>	

Zn

Legend and value threshold:

- ☒ Zinc (Zn)
- ☐ Normal
- ☐ Potential enrichment (Verify with EFZn)

label	rule
Normal	

Potential enrichment (Verify with EFZn)	
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SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>method</u>	" "
<u>geom</u>	

RC_{Zn} Zinc Relative Content

Legend and value threshold:

☒ Zinc (Zn) zinc relative content
☐ Zinc below natural geochemical relative range
☐ Zinc natural geochemical relative range
☐ Zinc above natural geochemical relative range

label	rule
Zinc below natural geochemical relative range	
Zinc natural geochemical relative range	
Zinc above natural geochemical relative range	

SIS code name

Short abstract

Measure unit %

Category

Involved SIS tables

Formula

$$RC_{Zn} = \frac{zn}{[cu + zn + pb]} * 100$$

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	

	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" " "
<u>geom</u>	

Water Infiltration Potential

Water Infiltration Potential

Legend and value threshold:

- ☒ Water Infiltration Potential
- ☐ Low layer infiltration potential
- ☐ Good layer infiltration potential (No threshold for over infiltration risk)

label	rule
Low layer infiltration potential	

Good layer infiltration potential (No threshold for over infiltration risk)

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" "
<u>project</u>	
<u>horizon</u>	
<u>value</u>	
<u>unit</u>	" "
<u>geom</u>	

Rupture resistance cemented soil

Legend and value threshold:

Legend
☒ Rupture resistance cemented soil
☐ Not cemented (NOC) - Potentially sensitive to wind erosion
☐ Extremely weakly, very weakly and weakly cemented (EWC, VWC and WEC) - Moderate potential for wind erosion
☐ Moderately cemented and Strongly cemented (MOC and STC) - Low potential for wind erosion
☐ Very strongly cemented (VSC) and Extremely strongly cemented (EXC) - Very low potential for wind erosion

label	rule
Not cemented - Potentially sensitive to wind erosion	
Extremely weakly, very weakly and weakly cemented - Moderate potential for wind erosion	
Moderately cemented and Strongly cemented - Low potential for wind erosion	
Very strongly cemented and Extremely strongly cemented - Very low potential for wind erosion	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " "
<u>date</u>	

<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>unit</u>	" " "
<u>rupture</u>	" " "
<u>geom</u>	

Texture

Legend and value threshold:

☒ texture
☐ Clay and Silty clay - Very slow infiltration rate
☐ Sandy clay loam and sandy clay - Slow infiltration rate
☐ Silty loam, loam, silty clay loam, silty clay loam, silty clay loam - Moderate infiltration rate
☐ Sand and loamy sand - High infiltration rate

label	rule
Clay and Silty clay - Very slow infiltration rate (C)	

Sandy clay loam and sandy clay - Slow infiltration rate (SCL, SC)	
Sandy loam, loam, silt loam, clay loam, silty clay loam, silt - Moderate infiltration rate (SL, L, SiL, CL, SiCL, Si)	
Sand and loamy sand - High infiltration rate (S, LS)	

SIS code name

Short abstract

Measure unit

Category

Involved SIS tables

Indicator table fields

name	description
<u>id</u>	
<u>point_id</u>	
<u>point_type</u>	" " "
<u>date</u>	
<u>upper</u>	
<u>lower</u>	
<u>survey_m_id</u>	" " " "
<u>project</u>	
<u>horizon</u>	
<u>unit</u>	" " "
<u>texture</u>	

	" " "
<u>geom</u>	

Bibliography
